

Announcements

- ▶ Any questions/issues?

- ▶ Schedule:

Today (Follow-up): Milankovitch cycles and observation in paleodata

Today (New): “Climate sensitivity” — how precisely does additional CO₂ cause warming

Tomorrow (Lab): Absorption by CO₂.

Thursday (Discussion): Absorption line of CO₂ and temperature increase.

Next Week: Timescales of climate change.

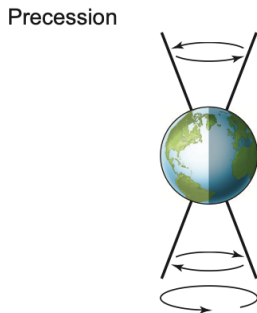
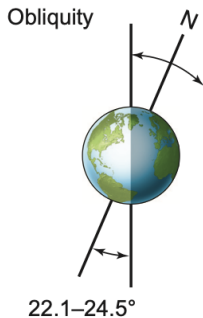
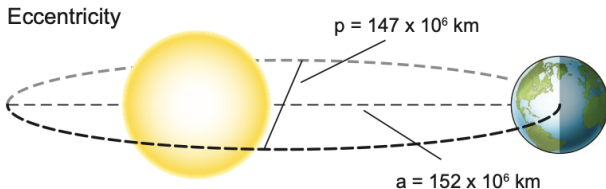
Today's Reading Quiz (3 min)

On your own sheet of paper. . .

Briefly explain saturation in terms of the “murky pond.” Then say a few words about saturation in the absorption lines of CO₂ and/or methane.

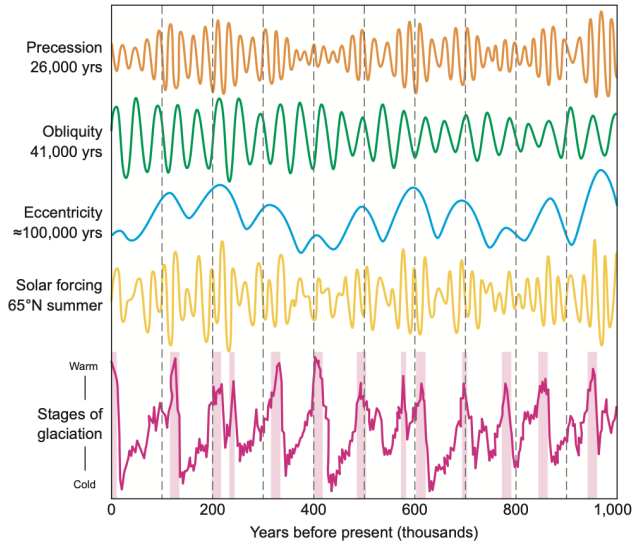
Turn in to Blackboard as scanned pdf (or image is fine).

Orbital effects on solar radiation

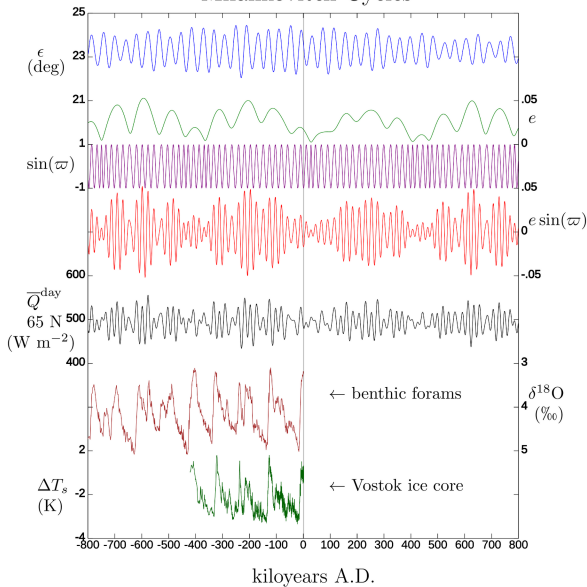


NASA videos showing each type of change

Orbital effects on solar radiation

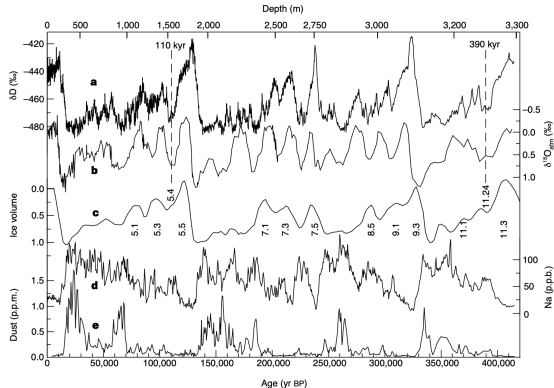


Milankovitch Cycles



Paleotemperature Data

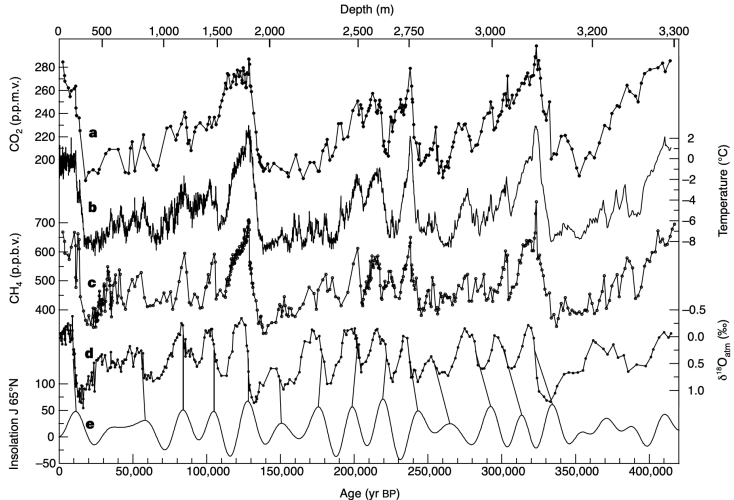
The “Vostok” ice cores from Antarctica reveals temperature proxy data and direct measurements of gas concentrations over 400,000 years from trapped bubbles.



$$\Delta T = \frac{\delta D - 8 \delta^{18}O}{9} \quad (\text{how temperature is estimated})$$

Paleotemperature Data

Overplotting the Milankovitch cycle prediction of solar flux makes some tantalizing (but not completely clear) connections.



Climate Sensitivity

The term **climate sensitivity** is often used by climate scientists in two ways:

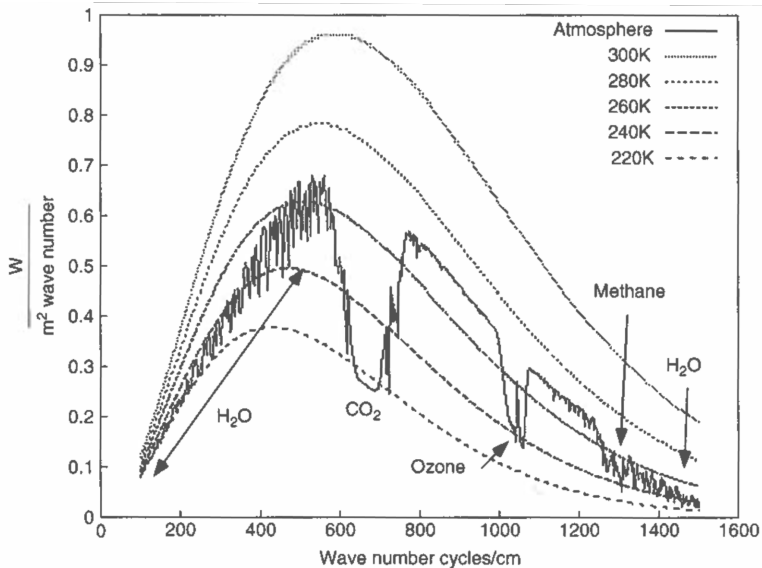
- ▶ The effect on temperature by a doubling of CO_2 in the atmosphere.
- ▶ The general connection between changes in CO_2 concentration (and other greenhouse gases) and the accompanying temperature rise.

This week we would like to explore exactly how additional CO_2 in the atmosphere, which causes additional *absorption* of the Earth surface's infrared blackbody to space, changes the equilibrium temperature of the Earth's surface.

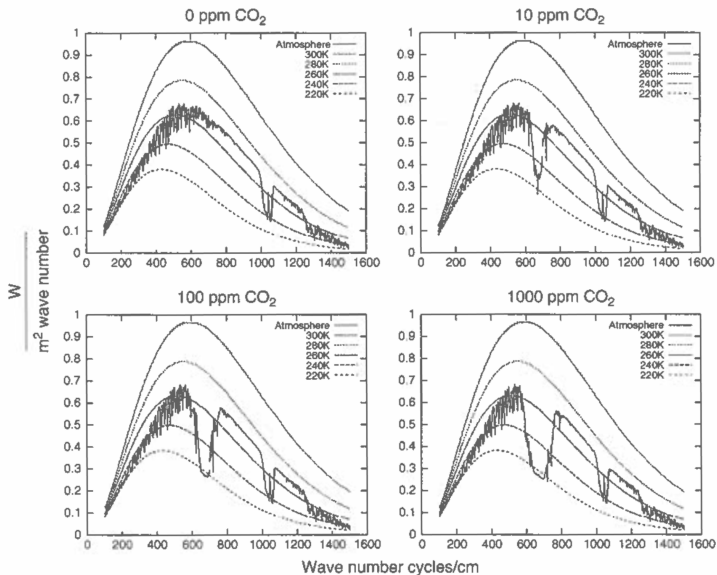
“Greenhouse Gases” reading (Archer, Chapter 4)

- ▶ A marked-up version of the chapter identifies key points of review — from our prior studies of blackbody radiation, absorption of infrared light due to changes in dipole moments, absorption lines in BB spectra, and the connection between temperature and altitude (the *lapse rate*) — and highlights of new information.
- ▶ Today in class we'll go quickly through the review points and then try to understand the key points that the figures convey.
- ▶ In Discussion on Thursday you will do some of this analysis yourself on your own blackbody spectrum curves.

Blackbody curves, absorption line saturation



Blackbody curves, increasing CO₂



Increasing CO₂ and lost power

